

Example: A manager decided to purchase 5000 shares when the stock was at \$10. All 5000 shares were transacted in the market, but at an average transaction price of \$10.50. If the commission fee was \$100 then implementation shortfall of the order is:

$$IS = 5000 \cdot (\$10.50 - \$10.00) + \$100 = \$2600$$

Opportunity Cost (Andre Perold)

The opportunity cost example refers to a situation where the manager does not transact the entire order. This could be due to prices becoming too expensive or simply a lack of market liquidity. Either way, it is essential that we account for all unexecuted shares in the implementation shortfall calculation. This process is as follows:

First, compute the paper portfolio return:

$$Paper\ Return = S \cdot P_n - S \cdot P_d$$

Next, compute the actual portfolio return for those shares that were executed:

$$Actual\ Return = \left(\sum s_j \right) P_n - \sum s_j p_j + fees$$

Then, the implementation shortfall is written as:

$$IS = (S \cdot P_n - S \cdot P_d) - \left(\left(\sum s_j \right) P_n - \sum s_j p_j + fees \right)$$

Let us now expand on this formulation. Share quantity S can be rewritten in terms of executed shares $\sum s_j$ and unexecuted shares $(S - \sum s_j)$ as follows:

$$S = \underbrace{\sum s_j}_{Executed} + \underbrace{\left(S - \sum s_j \right)}_{Unexecuted}$$

If we substitute the share quantity expression above into the previous IS formulation we have:

$$IS = \left(\sum s_j + \left(S - \sum s_j \right) \cdot P_n - \sum s_j + \left(S - \sum s_j \right) \cdot P_d \right) - \left(\left(\sum s_j \right) P_n - \sum s_j p_j + fees \right)$$

This equation can be written as:

$$IS = \underbrace{\sum s_j p_j - \sum s_j P_d}_{Execution\ Cost} + \underbrace{\left(S - \sum s_j \right) \cdot (P_n - P_d)}_{Opportunity\ Cost} + fees$$